

DMVPN Fase 2 IKEv1

VPN Hub and Spoke con DMVPN Fase 2 y Enrutamiento Dinámico

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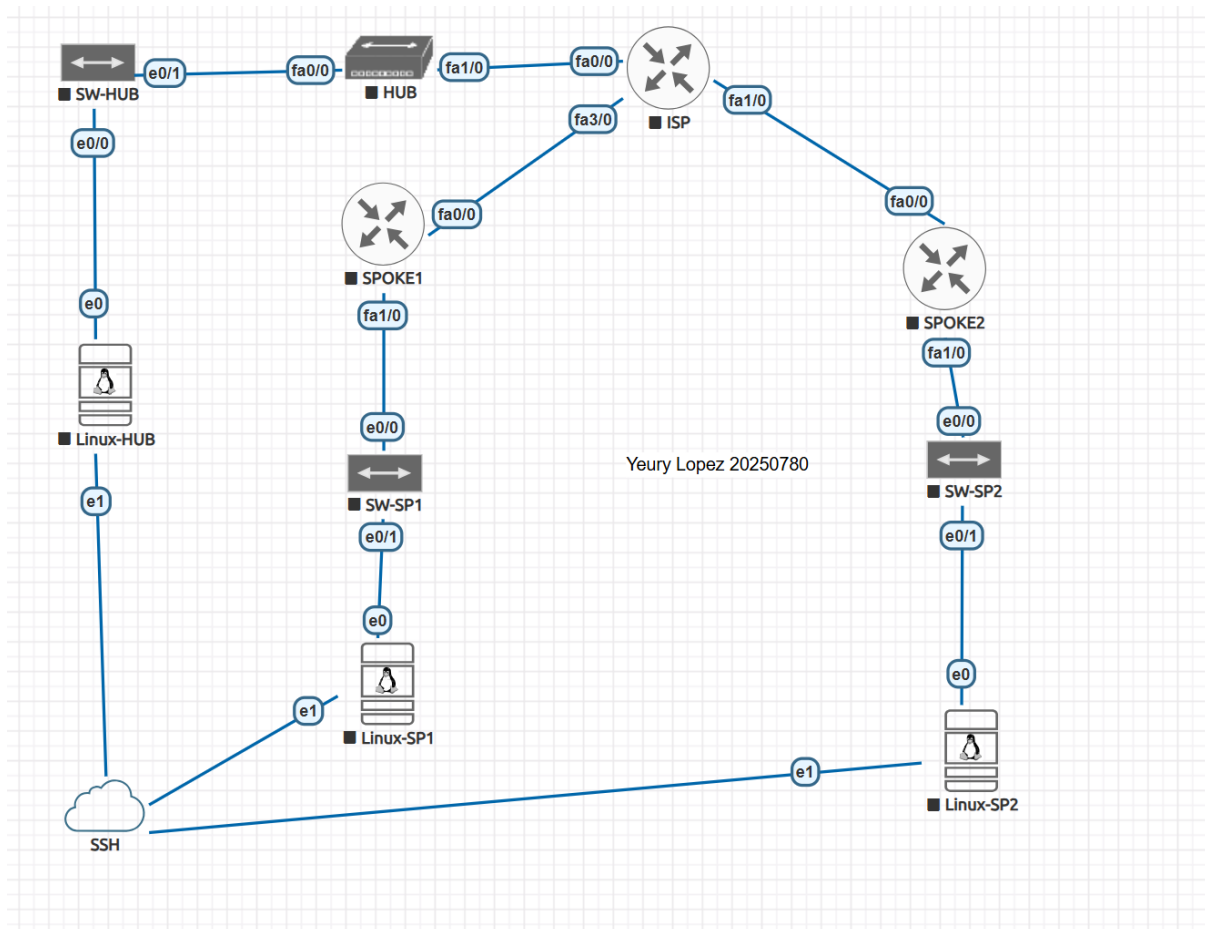
Entregable: P7

1. Objetivo

Configurar una VPN Hub and Spoke punto a multipunto DMVPN Fase 2 con IKEv1 y enrutamiento dinámico OSPF. DMVPN permite crear túneles VPN dinámicos entre múltiples sitios. En la Fase 2, los Spokes pueden comunicarse directamente entre sí sin pasar por el Hub, creando túneles dinámicos spoke-to-spoke mediante NHRP.

Topología de Red

La topología DMVPN consiste en un router HUB central y dos Spokes conectados a través de un router ISP. Cada router tiene una LAN con un host Linux para demostrar la conectividad.



Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Rol
HUB	Fa0/0	192.168.7.1/24	LAN
HUB	Fa1/0	10.7.82.1/30	WAN → ISP
ISP	Fa0/0	10.7.82.2/30	WAN → HUB
ISP	Fa3/0	10.7.83.1/30	WAN → SPOKE1
ISP	Fa1/0	10.7.84.1/30	WAN → SPOKE2
SPOKE1	Fa0/0	10.7.83.2/30	WAN → ISP
SPOKE1	Fa1/0	192.168.80.1/24	LAN
SPOKE2	Fa0/0	10.7.84.2/30	WAN → ISP
SPOKE2	Fa1/0	192.168.78.1/24	LAN
HUB Tunnel0	—	172.16.78.1/24	DMVPN
SPOKE1 Tunnel0	—	172.16.78.2/24	DMVPN
SPOKE2 Tunnel0	—	172.16.78.3/24	DMVPN

2. Parámetros VPN

Parámetro	Valor
Tipo de VPN	DMVPN Fase 2 con IKEv1
Protocolo Túnel	mGRE (Multipoint GRE)
Protocolo NHRP	NHRP Network-ID 780
Enrutamiento Dinámico	OSPF Proceso 1 — Area 0
Fase 1 — Cifrado	AES-128
Fase 1 — Hash	SHA-1
Fase 1 — Autenticación	Pre-Shared Key (PSK)
Fase 2 — Modo	Transport
Pre-Shared Key	Yeury0780
NHRP Holdtime	300 segundos
Hub NHS IP	172.16.78.1

4. Verificación y Funcionamiento

4.1 Estado DMVPN

```
HUB#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
  1 10.7.83.2      172.16.78.2    UP 00:15:17    D
  1 10.7.84.2      172.16.78.3    UP 00:15:15    D
```

El comando `show dmvpn` muestra los peers NHRP registrados. El tipo Hub confirma que este router es el servidor NHRP central. Los Spokes en estado UP confirman los túneles activos.

4.2 Estado OSPF

```
HUB#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.78.1      0    FULL/ -         00:01:38    172.16.78.3    Tunnel0
192.168.80.1      0    FULL/ -         00:01:47    172.16.78.2    Tunnel0
HUB#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst              src              state            conn-id status
10.7.82.1        10.7.83.2        QM_IDLE          1001 ACTIVE
10.7.82.1        10.7.84.2        QM_IDLE          1002 ACTIVE
IPv6 Crypto ISAKMP SA
```

Los vecinos OSPF en estado FULL confirman que el enrutamiento dinámico está funcionando correctamente a través del túnel DMVPN.

4.3 Estado ISAKMP

```
HUB#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst              src              state            conn-id status
10.7.82.1        10.7.83.2        QM_IDLE          1001 ACTIVE
10.7.82.1        10.7.84.2        QM_IDLE          1002 ACTIVE
IPv6 Crypto ISAKMP SA
```

Las sesiones IKEv1 activas confirman que el tráfico DMVPN viaja cifrado con IPsec.

4.4 Tabla de rutas

```
HUB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 10.7.82.2 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 10.7.82.2
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     10.7.82.0/30 is directly connected, FastEthernet1/0
L     10.7.82.1/32 is directly connected, FastEthernet1/0
     172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C     172.16.78.0/24 is directly connected, Tunnel0
L     172.16.78.1/32 is directly connected, Tunnel0
O     172.16.78.2/32 [110/1000] via 172.16.78.2, 00:14:09, Tunnel0
O     172.16.78.3/32 [110/1000] via 172.16.78.3, 00:13:59, Tunnel0
     192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
```

Las rutas OSPF en la tabla de enrutamiento confirman que el enrutamiento dinámico distribuye correctamente las rutas entre todos los sitios.

4.5 Demostración de conectividad

```
--- 192.168.78.2 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3035ms
rtt min/avg/max/mdev = 101.747/236.049/398.382/122.712 ms
PING 192.168.7.2 (192.168.7.2) 56(84) bytes of data:
64 bytes from 192.168.7.2: icmp_seq=1 ttl=62 time=361 ms
64 bytes from 192.168.7.2: icmp_seq=2 ttl=62 time=136 ms
64 bytes from 192.168.7.2: icmp_seq=3 ttl=62 time=538 ms
64 bytes from 192.168.7.2: icmp_seq=4 ttl=62 time=94.9 ms
```

El ping exitoso entre Spokes demuestra la conectividad spoke-to-spoke directa característica de DMVPN Fase 2.